# Optimization of multi-layer optical films with a memetic algorithm and mixed integer programming

Supporting Information

Yu Shi[1], Wei Li[1], Aaswath Raman[1], and Shanhui Fan[1, a)]

[1]*Department of Electrical Engineering, Ginzton Laboratory, Stanford University, Stanford, California 94305, USA*

[a)] *Author to whom correspondence should be addressed. Email: shanhui@stanford.edu.*

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**Part 1. Reselection and mutation rate analysis**

In this section, we examine the effects of the reselection rate, $r$, and mutation rate, $\mu$, in the memetic algorithm optimization for both the radiative cooling device and the incandescent light bulb filter. In both cases, we study the performance of the memetic algorithm with $r = [0.80, 0.90, 0.99]$ and $\mu = [0.01, 0.05, 0.10]$.

First, we perform the memetic algorithm optimization for the radiative cooling device. Aside from $r$ and $\mu$, all other optimization parameters are kept the same as those of the manuscript. For each combination of $r$ and $\mu$ listed above, we apply the memetic algorithm for 30 trials over 105 generations. In each generation, for statistical robustness, we plot the median and inter-quartile range of these trials' minimum merit functions (M.F.) in Fig. S1. As we can see, at a given $\mu$, a higher $r$ value generally speeds up the decrease in the merit function. For a small $\mu = 0.01$, a large $r = 0.99$ in general helps the algorithm to converge to a slightly more optimum structure. For a larger $\mu = 0.10$, a moderate $r = 0.90$ helps the algorithm to converge towards a slightly more optimum structure. Furthermore, a low $r = 0.80$ value produced relatively sub-optimal results but was more consistent, whereas a moderate $r = 0.90$ value and a high $r = 0.99$ value exhibit more spread in their resulting structures' merit functions. These observations could indicate that the search landscape of the radiative cooling merit function could have a few prominent local minima that are high-performing. A high reselection rate helps the population to retain individuals at these prominent local optima, whereas a low reselection rate can more likely lose individuals at the prominent local minima and steer the population into search regions with less optimal local minima.

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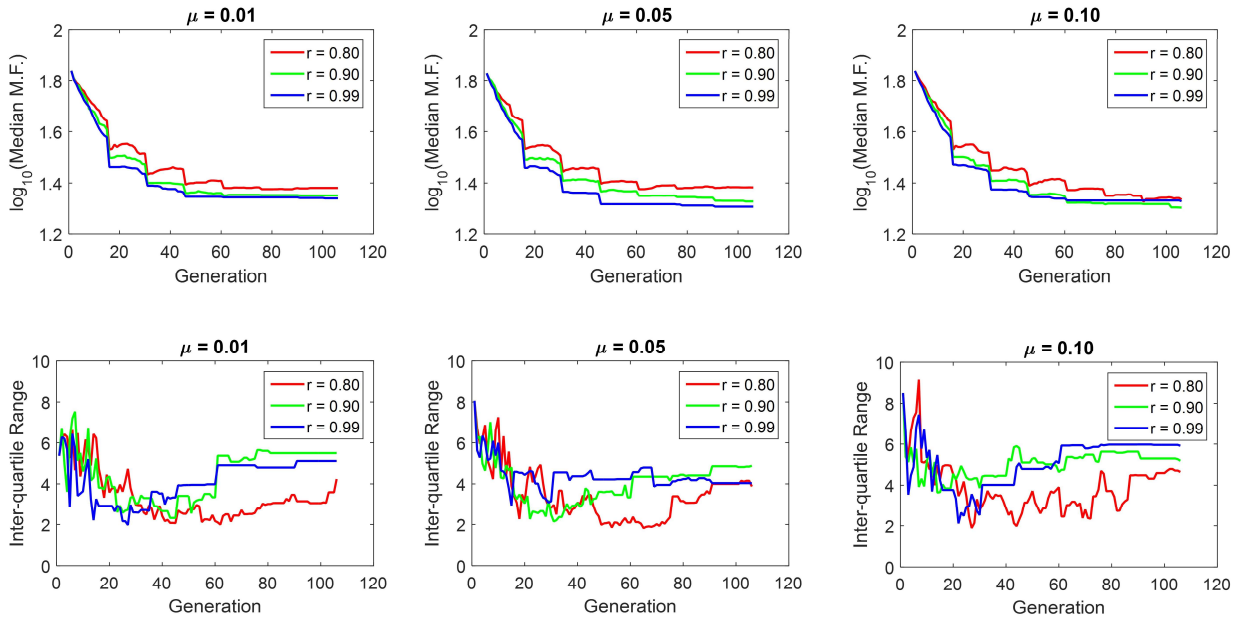



**Figure S1.** Statistics of the radiative cooling device optimization over 30 trials for varying $r$ and $\mu$. The plot shows the median (top row) and inter-quartile range (bottom row) of the merit function (M.F.) at each generation for $r = [0.80, 0.90, 0.99]$ and $\mu = [0.01, 0.05, 0.10]$.

Next, we perform the memetic algorithm optimization for the incandescent light bulb filter. As we did above, we keep all optimization parameters the same as those of the manuscript except for $r$ and $\mu$. For each combination of $r$ and $\mu$ above, we apply the memetic algorithm for 15 trials over 200 generations. In Fig. S2, we plot the median and inter-quartile range of the trials' minimum merit functions at each generation. Here, we notice that for a given $\mu$, in the initial generations, having a high $r$ value accelerated the decrease of the merit function. Nevertheless, the end-result of the optimization differed considerably. For a small $\mu = 0.01$, on average, all $r$ values converged to devices with roughly the same performance with similar consistency. On the other hand, for larger $\mu$ values of 0.05 and 0.10, a low reselection rate of $r = 0.80$ more consistently converged to more optimal devices than both a moderate $r = 0.90$ and a high $r = 0.99$. We can



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infer from these observations that the search landscape of the light bulb filter merit function contains much more local minima than the search landscape of the radiative cooling device, and a low reselection rate in this case helps the population examine more regions of the search landscape. The increase in local minima is expected because with the increase in the number of layers, the search landscape contains many more degrees of freedom than the landscape in the radiative cooling device optimization. In such a rugate search landscape, it is helpful to have a lower reselection rate $r$ and higher mutation rate $\mu$ to allow the population to diversify its exploration into many of the local minima.

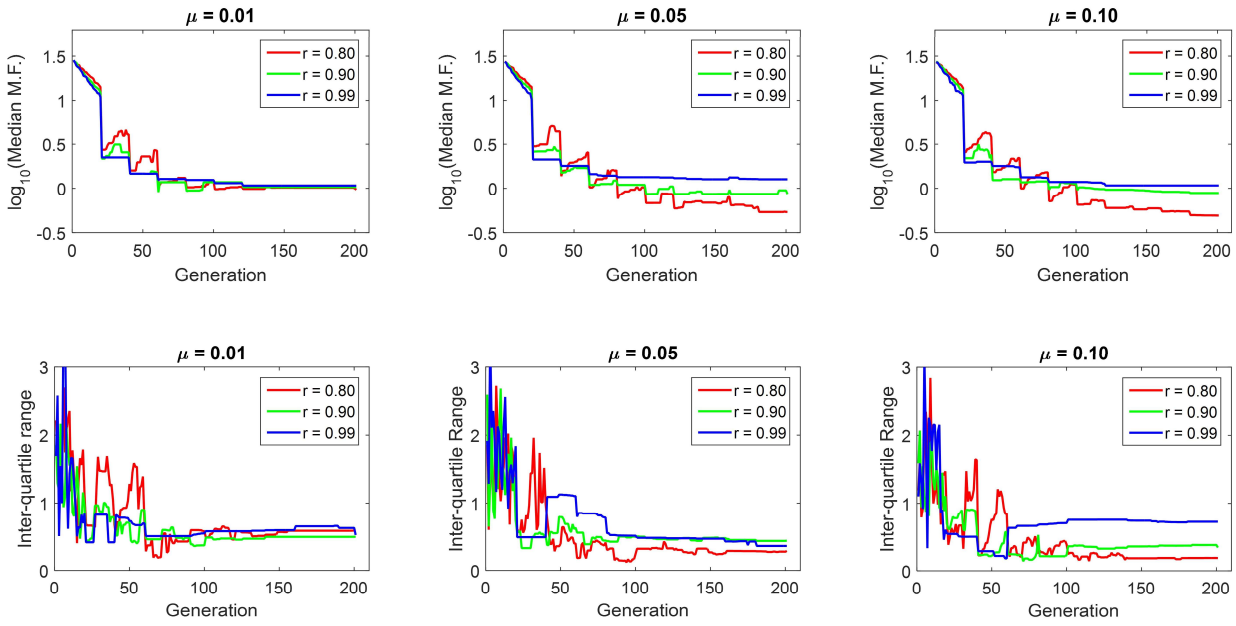



**Figure S2.** Statistics of the light bulb filter optimization over 15 trials for varying $r$ and $\mu$. The plot shows the median (top row) and inter-quartile range (bottom row) of the merit function (M.F.) at each generation for $r = [0.80, 0.90, 0.99]$ and $\mu = [0.01, 0.05, 0.10]$.



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Finally, we investigate the effect of having a highly mutating population. We fix $r = 0.90$, and we apply the memetic algorithm to both the radiative cooler and light bulb filter for 30 and 15 trials, respectively, with $\mu = [0.50, 0.80]$. In Fig. S3, we plot the median and inter-quartile range of these trials' minimum merit functions, along with the results from previous trials with $\mu = 0.10$ for comparison. For the radiative cooling device plots in Fig. S3(a), we notice that while all three mutation rates show similar consistency, a mutation rate of $\mu = 0.50$ on average out-performed mutation rates of $\mu = 0.10$ and $\mu = 0.80$. On the other hand, for the light bulb filter plots in Fig. S3(b), we observe that having a very high mutation rate of $\mu = 0.80$ on average helps with the convergence to more optimal structures, which is consistent with the speculation that the search landscape for the light bulb filter is rugate and could benefit from high mutation that diversifies the search regions.

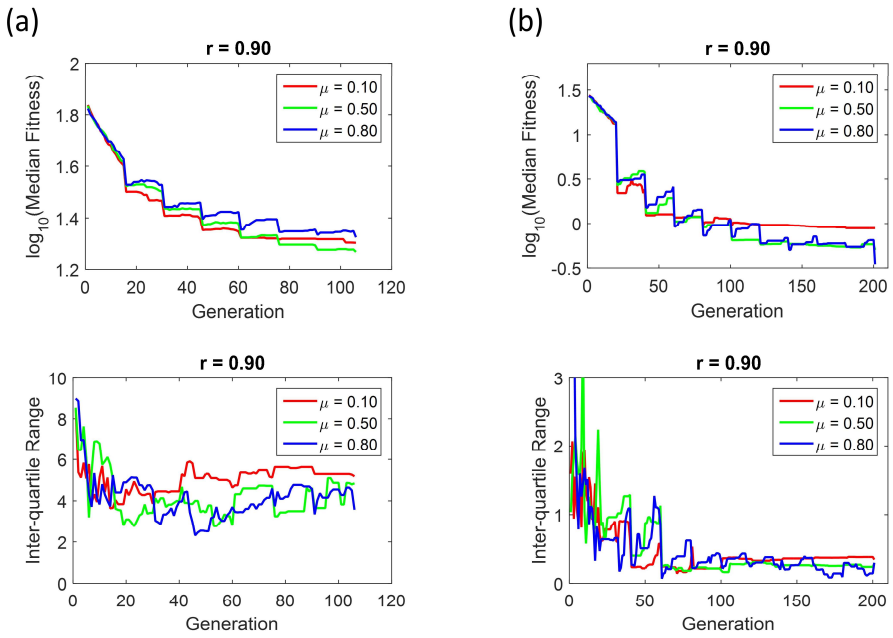



**Figure S3.** Statistics of the merit function for (a) the radiative cooling device optimization and (b) the light bulb filter optimization with high mutation rates. The reselection rate is fixed at $r = 0.90$, and the mutation rates are $\mu = [0.10, 0.50, 0.80]$.



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In summary, from the examples above, we see that while a high $r$ value in general accelerates the decrease in merit function in the initial generations, the optimum choice of $\mu$ and $r$ are problem specific depending on the search landscape of the merit function. Search landscapes with a few prominent local optima favor the use of moderate reselection and mutation rates, whereas rugate search landscapes with many local optima favors the use of low reselection and high mutation rates in order to explore more regions with local optima. In such evolutionary algorithm, it is generally important to diversify the population in order to explore more local minimum regions.

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**Part 1. Radiative cooling device and light bulb filter optimization with two materials**

In this section, we present the memetic optimization results for the spectra of both the radiative cooling device[1] and incandescent light bulb filter[2] with the same material composition as those reported in Refs. 1 and 2, respectively. In applying the memetic algorithm with such fixed material arrangements, we modify the "generating the population" step such that all starting individuals have the same material composition as Refs. 1 and 2, but with random thicknesses. We then proceed with the "crossover", "mutation", "reselection of population", and "refinement of elite individuals" steps identical to the descriptions in the manuscript. The results below demonstrate that while fixing the material's arrangements, our memetic algorithm can effectively identify better performing structures than those reported in Refs. 1 and 2. Compared to the optimized spectra in the manuscript, however, since the structures below are inferior to those described in the manuscript, these optimizations also show that it is advantageous to use a combination of many materials to reduce the number of material layers.

In the optimization of the radiative cooling device, we arrange seven layers of $SiO_2$ and $HfO_2$ placed on top of a silver substrate as was done in Ref. 1. We then apply the memetic algorithm to optimize the thicknesses of each layer according to the target spectrum for the optimization is shown in Fig. S4(a). The resulting structure's layer thicknesses are shown in Table S1, and the structure's angular dependent emissivity is plotted in Fig. S4(b). In addition, the structure's normal and angle-averaged emissivity/absorptivity are plotted in Fig. S4(a), which show low solar absorptivity and high emissivity in atmospheric transparency window as expected. Using the same solar and atmospheric conditions as Ref. 1, we compute that the optimized structure in Table S1 could reach an equilibrium temperature of 20.8 K below ambient in the presence of direct sunlight. This value exceeds the theoretical predictions in Ref. 1 of 19.5 K below



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ambient, but it is not as low as the six-layer structure with 3 materials discussed in the manuscript, which could reach 23.4 K below ambient temperature.

**Table S1.** Material composition and thicknesses for the optimized radiative cooler using 2 materials

| Layer # | Material | Thickness (μm) |
|---|---|---|
| | Air | Superstrate |
| 1 | $SiO_2$ | 214 |
| 2 | $HfO_2$ | 681 |
| 3 | $SiO_2$ | 779 |
| 4 | $HfO_2$ | 127 |
| 5 | $SiO_2$ | 61 |
| 6 | $HfO_2$ | 47 |
| 7 | $SiO_2$ | 50 |
| | Ag | Substrate |

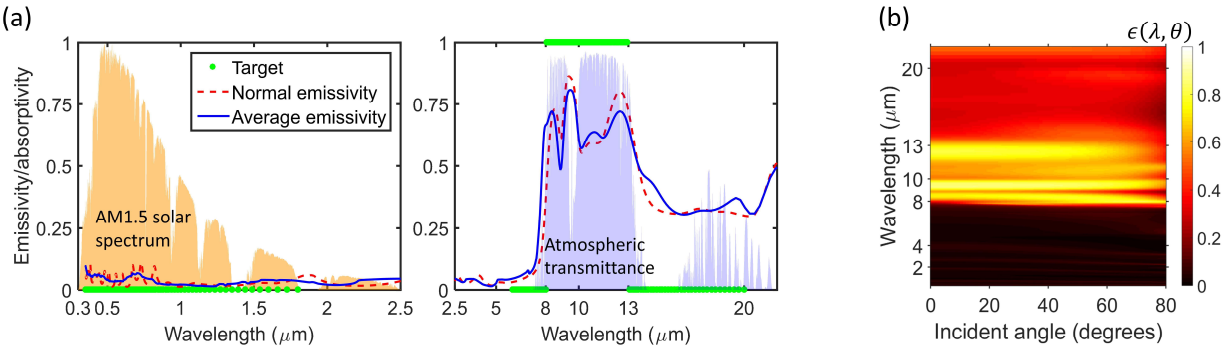



**Figure S4.** Emissivity/absorptivity spectrum of the seven-layer structure using $SiO_2$ and $HfO_2$ shown in Table S1. (a) Plot of the target, normally-incident, and angle-averaged emissivity/absorptivity spectrum. (b) Plot of the emissivity/absorptivity's angle dependence.

In the optimization of the incandescent light bulb filter, we arrange 90 layers of $Ta_2O_5$ and $SiO_2$ surrounded by air, as was the case in Ref. 2. The target reflection spectrum for this



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optimization is shown in Fig. S5(a), and we perform the memetic algorithm to optimize on the layers' thicknesses. The algorithm removed 10 total layers to produce an 80-layer structure that is shown in Table S2. We compute and plot this structure's angle-dependent reflectivity in Fig. S5(b), and we show the normal-direction and angle-averaged reflectivity of this structure in Fig. S5(a). As anticipated, this filter exhibits high transmission in the visible range, but high reflection in ultraviolet and infrared such that it could help recycle non-visible incandescent light bulb radiation as to further heat up the light bulb emitter. As was done in the manuscript, we compute that the temperature this emitter can reach with 100 W of input power is 3677 K, and we plot its spectral emissive power at that temperature is shown in Fig. S5(c). From Fig. S5(c), we compute using Eqn. (7) that this optimized 80-layer structure can attain a spectral enhancement factor of $\chi = 14.1$. This is higher than the emissivity enhancement factor calculated with the 90-layer structure in Ref. 2 of $\chi = 12.8$, and it is slightly less than the optimized 41-layer structure using 7 materials that has $\chi = 14.8$ shown in the manuscript.

With the two optimization examples shown above, we have demonstrated that the memetic algorithm can effectively produce structures with the same material composition as those used in the literature but with superior performance. Compared to the optimization examples in the manuscript that have fewer material layers but better performance, it is evident that in terms of reducing the layers of materials, it is beneficial to design the multi-layer structure from a diverse set of materials.

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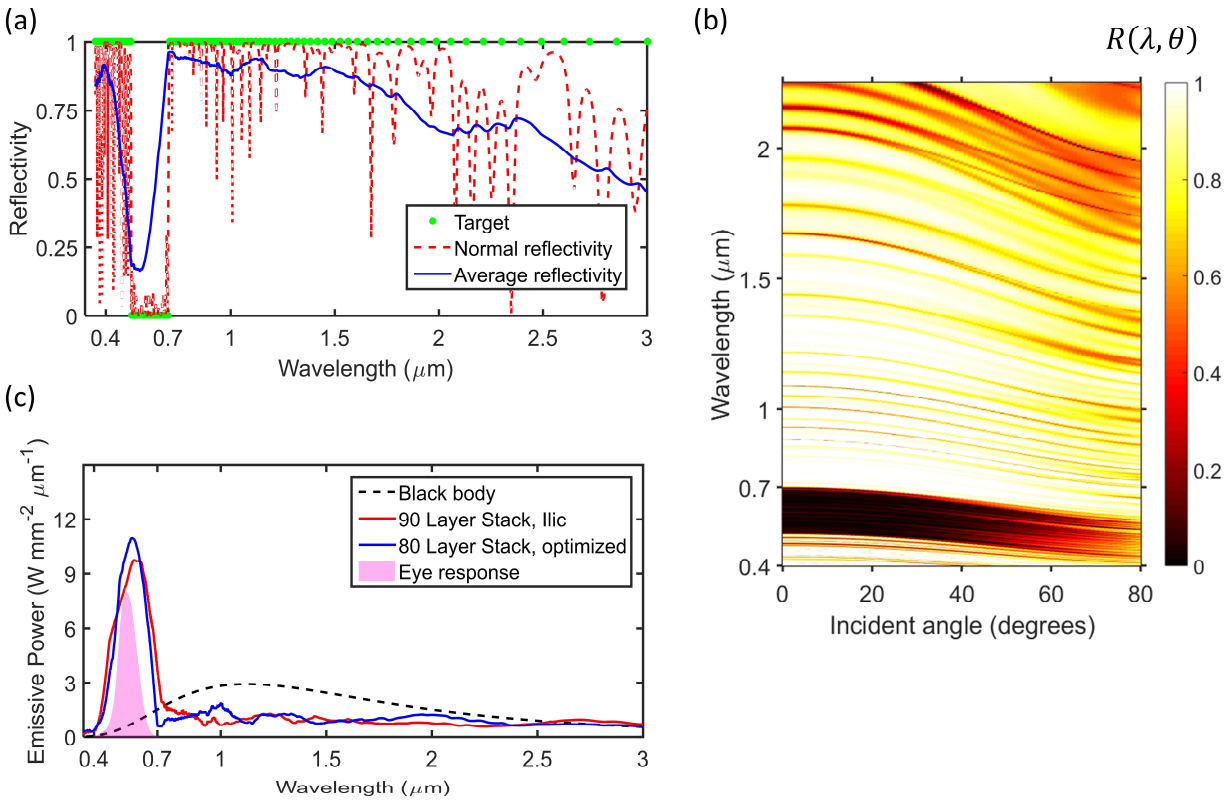



**Figure S5**. (a) The reflectivity spectrum of the 80-layer incandescent light bulb filter using $SiO_2$ and $Ta_2O_5$. The green dots are the target reflectivity spectrum in the normal direction. The red dotted line and the blue solid line are the normal and angle-averaged reflectivity spectrum of the structure shown in Table S2, respectively. (b) Angle-dependence of the reflectivity spectrum for the structure shown in Table S2. (c) Comparison between the structure in Table S2 (blue) and Ref. 2 (red) given the same input power of 100 W to the emitter. The eye's responsivity and a black body's emissive power spectrum at the same input power are plotted for reference.

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**Table S2**. Material composition and thicknesses for the optimized light bulb filter using 2 materials

| Layer # | Material | Thickness (nm) | Layer # | Material | Thickness (nm) | Layer # | Material | Thickness (nm) |
|---|---|---|---|---|---|---|---|---|
| | Air | - | 31 | $Ta_2O_5$ | 115 | 62 | $SiO_2$ | 254 |
| 1 | $Ta_2O_5$ | 3 | 32 | $SiO_2$ | 147 | 63 | $Ta_2O_5$ | 159 |
| 2 | $SiO_2$ | 72 | 33 | $Ta_2O_5$ | 10 | 64 | $SiO_2$ | 239 |
| 3 | $Ta_2O_5$ | 248 | 34 | $SiO_2$ | 202 | 65 | $Ta_2O_5$ | 284 |
| 4 | $SiO_2$ | 396 | 35 | $Ta_2O_5$ | 101 | 66 | $SiO_2$ | 207 |
| 5 | $Ta_2O_5$ | 388 | 36 | $SiO_2$ | 169 | 67 | $Ta_2O_5$ | 160 |
| 6 | $SiO_2$ | 183 | 37 | $Ta_2O_5$ | 248 | 68 | $SiO_2$ | 204 |
| 7 | $Ta_2O_5$ | 242 | 38 | $SiO_2$ | 196 | 69 | $Ta_2O_5$ | 259 |
| 8 | $SiO_2$ | 174 | 39 | $Ta_2O_5$ | 388 | 70 | $SiO_2$ | 207 |
| 9 | $Ta_2O_5$ | 127 | 40 | $SiO_2$ | 191 | 71 | $Ta_2O_5$ | 299 |
| 10 | $SiO_2$ | 201 | 41 | $Ta_2O_5$ | 105 | 72 | $SiO_2$ | 207 |
| 11 | $Ta_2O_5$ | 105 | 42 | $SiO_2$ | 23 | 73 | $Ta_2O_5$ | 237 |
| 12 | $SiO_2$ | 159 | 43 | $Ta_2O_5$ | 151 | 74 | $SiO_2$ | 197 |
| 13 | $Ta_2O_5$ | 117 | 44 | $SiO_2$ | 162 | 75 | $Ta_2O_5$ | 217 |
| 14 | $SiO_2$ | 55 | 45 | $Ta_2O_5$ | 22 | 76 | $SiO_2$ | 210 |
| 15 | $Ta_2O_5$ | 126 | 46 | $SiO_2$ | 221 | 77 | $Ta_2O_5$ | 60 |
| 16 | $SiO_2$ | 150 | 47 | $Ta_2O_5$ | 218 | 78 | $SiO_2$ | 220 |
| 17 | $Ta_2O_5$ | 251 | 48 | $SiO_2$ | 213 | 79 | $Ta_2O_5$ | 300 |
| 18 | $SiO_2$ | 224 | 49 | $Ta_2O_5$ | 23 | 80 | $SiO_2$ | 95 |
| 19 | $Ta_2O_5$ | 157 | 50 | $SiO_2$ | 167 | | Air | |
| 20 | $SiO_2$ | 81 | 51 | $Ta_2O_5$ | 105 | | | |
| 21 | $Ta_2O_5$ | 133 | 52 | $SiO_2$ | 169 | | | |
| 22 | $SiO_2$ | 128 | 53 | $Ta_2O_5$ | 236 | | | |
| 23 | $Ta_2O_5$ | 124 | 54 | $SiO_2$ | 217 | | | |
| 24 | $SiO_2$ | 249 | 55 | $Ta_2O_5$ | 42 | | | |
| 25 | $Ta_2O_5$ | 142 | 56 | $SiO_2$ | 220 | | | |
| 26 | $SiO_2$ | 104 | 57 | $Ta_2O_5$ | 240 | | | |
| 27 | $Ta_2O_5$ | 145 | 58 | $SiO_2$ | 182 | | | |
| 28 | $SiO_2$ | 267 | 59 | $Ta_2O_5$ | 92 | | | |
| 29 | $Ta_2O_5$ | 266 | 60 | $SiO_2$ | 160 | | | |
| 30 | $SiO_2$ | 145 | 61 | $Ta_2O_5$ | 128 | | | |

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(2) Ilic, O.; Bermel, P.; Chen, G.; Joannopoulos, J. D.; Celanovic, I.; Soljačić, M. Tailoring High-Temperature Radiation and the Resurrection of the Incandescent Source. *Nat. Nanotechnol.* **2016**, *11* (4), 320–324.